Architecture-Independent Routing Reorganization in Vision-Language-Action Models

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Abstract

Vision-Language-Action models (VLAs) fine-tune large language models on robot demonstration data, yet how training reshapes the internal routing of instruction information at the token level—and whether these effects generalize across architectures—remains uncharacterized. We present a systematic token-level analysis framework and apply it to three VLA families built on distinct backbones (Llama-2-7B, Phi-3-mini, Qwen2.5-VL-3B) across four LIBERO task suites (10 architecture×suite cells). We identify two *architecture-independent* training signatures: (1) **R^2 exit collapse**—task-finetuned VLAs show a consistent drop in action-probe R^2 from mid-layer peak to exit layer (trained Δ=0.12–0.69 vs. untrained Δ=0.01–0.08), confirmed in 9 of 10 cells; and (2) **heightened counterfactual sensitivity**—trained models exhibit 7–10× greater sensitivity to instruction perturbations than their untrained counterparts. A three-tier routing gradient emerges: VLM base models show no action routing, general-purpose VLA pretraining establishes routing without exit collapse, and task-specific fine-tuning adds exit collapse with task-specialized counterfactual patterns. We further distinguish training-induced effects (exit collapse, counterfactual sensitivity) from base-model-dependent effects (L1 attention head counts), and observe that, under RSA, vision encoder geometry follows an inverted-U profile across LLM layers with functional probes confirming an L8–L13 decodability peak. Our framework and findings provide mechanistic grounding for VLA architecture design and training diagnostics.

1 Introduction

Vision-Language-Action models (VLAs) repurpose pretrained vision-language models for robot control by fine-tuning on demonstration data [2, 5]. While behavioral evaluations reveal that VLAs underutilize language instructions [3, 4], the *internal* mechanisms by which training reshapes instruction routing remain poorly understood. Prior work has not decomposed the representational roles of different token positions, explained how fine-tuning reorganizes information flow, or tested whether such effects generalize beyond a single VLA architecture.

We address these gaps with three contributions:

1. **A systematic token-level analysis framework** applicable to any decoder-only VLA. The framework combines cross-instruction probing, per-layer attention profiling, representational similarity analysis (RSA), and counterfactual perturbation, validated on 3 architectures × 4 task suites (10 cells with full data out of 12 possible).

2. **Architecture-independent training signatures.** Across OpenVLA (Llama-2-7B), TracVLA (Phi-3-mini), and NORA (Qwen2.5-VL-3B), we find that task-specific fine-tuning consistently produces (a) R^2 exit collapse (mean trained Δ=0.226, 9/10 cells confirmed) and (b) heightened counterfactual sensitivity (7–10× amplification). We distinguish these *training-induced* effects from *base-model-dependent* effects such as L1 attention head counts, which reflect VLM/LLM pretraining rather than VLA fine-tuning.

3. **A three-tier routing gradient.** Routing *emerges* at the VLM→VLA transition (general-purpose pretraining establishes instruction routing without exit collapse) and *reshapes* at the general→task-specific transition (task fine-tuning adds exit collapse and task-specialized sensitivity).

As a qualified observation, we report that under RSA, vision encoder geometry follows an inverted-U profile across LLM layers (peak at L12, ρ=0.628), though this finding is metric-dependent: CKA yields a reversed ordering. Functional probes independently confirm a mid-layer decodability peak (L8–L13).

2 Related Work

Vision-Language-Action Models. VLAs [2, 5] fine-tune vision-language models to predict robot actions conditioned on visual observations and language instructions. Recent architectures include OpenVLA [5] (Llama-2-7B backbone), TracVLA (Phi-3-mini backbone), and NORA (Qwen2.5-VL-3B backbone with grouped-query attention). While task performance has improved steadily, the internal mechanisms by which these models integrate multimodal information for action generation are largely unexplored.

Instruction Following in Robotics. Grant *et al.* [3] demonstrated visual dominance in VLA action generation, and Haon *et al.* [4] identified instruction underutilization across multiple VLA families. These behavioral studies establish the phenomenon but do not provide mechanistic explanations at the representational level.

Representational Analysis of Transformers. Probing classifiers [1], representational similarity analysis [7], and centered kernel alignment [6] are standard tools for analyzing transformer internals. We adapt these to the multimodal, autoregressive VLA setting where image, instruction, and action tokens occupy distinct positions and roles within a shared sequence.

3 Method: Token-Level Analysis Framework

We propose a framework for systematically analyzing how VLAs process and route instruction information at the token level. The framework operates on decoder-only VLA architectures where the input sequence consists of image tokens, instruction tokens, and action tokens, processed autoregressively.

3.1 Factorial Design

For each architecture×suite cell, we compare three conditions:

- **Trained**: task-specific fine-tuned VLA checkpoint.
- **Untrained (OXE-pretrained or VLM base)**: the model before task-specific fine-tuning—either an OXE-pretrained VLA (Tier 2) or a raw VLM (Tier 1).
- **Instruction ablation**: removing or perturbing instruction tokens (empty, wrong task, shuffled, paraphrased).

3.2 Analysis Methods

(1) Cross-Instruction Probing. Given the same visual frame paired with K distinct instructions, we train a K-way classifier on per-layer hidden representations. We extract representations at two token positions: `image_mean` (mean-pooled image tokens) and `last_preaction` (last token before action generation). Layer-wise accuracy profiles reveal where and how instruction identity becomes decodable.

(2) Action Probe (R^2). A linear probe maps per-layer representations to ground-truth actions. Per-layer R^2 profiles characterize how action-relevant information distributes across depth, and the difference between peak and exit R^2 quantifies *exit collapse*.

(3) Attention Profiling. We compute per-head attention fractions to instruction vs. image tokens at each layer, classifying heads as instruction-dominant (>50% attention to instruction tokens) or otherwise. This reveals the layer-wise attention routing pattern.

Table 1: Cross-instruction probe accuracy (%) at `last_preaction` position (OpenVLA, LIBERO-Goal, 10-way classification). LOOCV: trained 100.00±0.00 (L8–L31); untrained decays from 96.64±7.07 to 84.76±11.59.

Condition	L0	L1	L2	L5	L16	L24	L31
Trained	10.0	99.6	100	100	100	100	100
Llama-2 base	10.0	80.2	98.7	100	100	100	100
Untrained (rand)	10.0	100	100	99.x	92.3	86.4	82.0
Trained `img_mean`	10.0	—	—	—	10.0	10.0	9.6

(4) Representational Similarity Analysis (RSA). We compute RSA between VLA layer representations and action-aligned reference representations (from standalone encoders), measuring geometric alignment across depth.

(5) Counterfactual Perturbation. We systematically replace instructions (empty, wrong task, shuffled tokens, paraphrased) and measure the resulting shift in hidden representations (Cohen's d for trained models, cosine distance for untrained models that cannot generate meaningful actions).

(6) CKA Verification. Centered kernel alignment provides a complementary similarity metric to RSA, allowing us to assess metric robustness of geometric findings.

3.3 Architectures and Datasets

We evaluate three VLA families spanning different LLM backbones and attention mechanisms:

- **OpenVLA**: SigLIP+DINOv2 vision encoder, Llama-2-7B backbone (32 layers, multi-head attention).
- **TracVLA**: Phi-3-mini backbone (32 layers, multi-head attention).
- **NORA**: Qwen2.5-VL-3B backbone (36 layers, grouped-query attention, 16 heads / 2 KV heads).

All are evaluated on LIBERO benchmark suites: Goal, Spatial, Object, and Long (where checkpoints are available), yielding a 3×4 matrix with 10 of 12 cells populated (NORA lacks Object and Long fine-tuned checkpoints).

4 Experiments

4.1 Token-Level Specialization (Framework Validation)

Image tokens are instruction-invariant. Across all conditions and all 32 layers, cross-instruction probe accuracy on `image_mean` tokens remains at chance (10% for 10-way classification). This confirms that, as expected from the causal mask, image tokens carry no instruction-specific information—but importantly, this holds even though image and instruction tokens share a residual stream with learned LLM weights.

Three qualitatively different routing profiles. Table 1 shows the cross-instruction probe accuracy at `last_preaction` across layers for three conditions on OpenVLA/LIBERO-Goal. The trained model achieves near-perfect accuracy from L1 onward (99.6% at L1, 100% from L2). The Llama-2 pretrained base reaches 80.2% at L1, saturating by L5. The untrained (random init) model starts at 100% at L1 but decays to 82% by L31—instruction signal degrades through random weights.

Six-method systematic validation. Table 2 shows that all six probing methods agree on the ordering trained ≈ untrained ≫ DINO across two benchmarks, confirming framework robustness.

Table 2: Six probing methods on two LIBERO benchmarks: consistent ordering trained ≈ untrained ≫ DINO. Best per-method scores shown.

	Trained Best	Untrained Best	DINO	Ordering
Temporal ρ	0.955	0.933	0.901	T ≈ U ≫ D
Action R^2	0.683	0.617	0.687	T ≈ U ≫ D

Table 3: R^2 exit collapse matrix: trained Δ (peak − exit) across 3 architectures × 4 LIBERO suites. ✓: $\Delta > 0.08$; △: OFT outlier. Mean untrained $\Delta = 0.032$.

Architecture (Backbone)	Goal	Spatial	Object	Long	Mean
OpenVLA (Llama-2-7B)	0.150 ✓	0.130 ✓	0.121 ✓	0.004 △	0.101
TracVLA (Phi-3-mini)	0.228 ✓	0.138 ✓	0.294 ✓	0.198 ✓	0.215
NORA (Qwen2.5-VL-3B)	0.691 ✓	0.302 ✓	—	—	0.497
Column Mean	0.356	0.190	0.208	0.101	**0.226**

4.2 Architecture-Independent Routing Reorganization

4.2.1 R^2 Exit Collapse

Table 3 presents the complete 3×4 R^2 exit collapse matrix. For each cell, we report the difference Δ between peak and exit R^2 at the `last_preaction` position. Task-finetuned models show $\Delta > 0.08$ in 9 of 10 cells (mean Δ=0.226, median=0.168). Untrained models show negligible collapse (mean Δ=0.032).

OpenVLA exit collapse detail. Table 4 shows per-condition R^2 profiles for OpenVLA on Goal and Spatial suites. Trained models exhibit clear inverted-U profiles: peak at L13–L14, exit R^2 dropping to 0.528–0.540. OXE-pretrained (untrained) models maintain flat profiles with minimal collapse (Δ=0.016–0.020), despite higher absolute R^2 values (0.825–0.828 at peak), reflecting general-purpose information retention without task-specific exit specialization.

Full-layer R^2 scan. On LIBERO-Goal alone, we perform a dense 33-layer × 2-condition × 2-aggregation scan (132 probes; Table 5). The trained `image_mean` peak occurs at L8 (R^2=0.655); trained `last_preaction` peaks at L13 (R^2=0.678). Untrained models show peak at L0 for `image_mean` (0.640) and L32 for `last_preaction` (0.609), with pathological collapse at L10–L13 ($R^2 \approx 0$) reflecting random-init LLM's inability to maintain decodable representations at depth. Best-vs-best comparison: `image_mean` Δ=0.015 (p=0.006), `last_preaction` Δ=0.069 (p¡0.0001).

Cross-benchmark replication. On LIBERO-Spatial (Table 6), the trained inverted-U replicates with peaks shifted 2–8 layers deeper (L13 for `image_mean`, L21 for `last_preaction`), possibly reflecting deeper processing requirements for spatial reasoning. RSA confirms the pattern: trained peak at L14 (ρ=0.691) vs. untrained peak at L0 (ρ=0.652).

OFT outlier. The sole exception to exit collapse is OpenVLA Long (Δ=0.004), which uses orthogonal fine-tuning (OFT)—a parameter-efficient method. TracVLA Long with full fine-tuning shows Δ=0.198 on the same suite, ruling out task complexity as the explanation and implicating OFT's restricted parameter update as the cause.

4.2.2 Counterfactual Sensitivity

Table 7 shows counterfactual sensitivity for trained models. Across architectures, the ordering is consistently empty > wrong > shuffled > paraphrased, with "empty" ranking first in 11 of 12 trained settings. Cohen's d ranges from 0.70 to 1.52, indicating medium-to-large effects.

Table 4: OpenVLA R^2 exit collapse on Goal and Spatial suites (`last_preaction`).

Suite	Condition	Peak R^2	Peak Layer	Exit R^2 (L31)	Δ
Goal	Trained (LIBERO-finetuned)	0.678	L13	0.528	0.150
	Untrained (OXE pretrained)	0.825	L21	0.805	0.020
Spatial	Trained (Spatial-finetuned)	0.670	L14	0.540	0.130
	Untrained (OXE pretrained)	0.828	L22	0.812	0.016

Table 5: Full-layer R^2 scan (OpenVLA, LIBERO-Goal). Peak and exit layer R^2 values.

Condition	Readout	Peak Layer	Peak R^2	Exit R^2	Δ
Trained	`image_mean`	L8	0.655	$\sim$0.45	$\sim$0.20
Trained	`last_preaction`	L13	0.678	$\sim$0.55	$\sim$0.13
Untrained	`image_mean`	L0	0.640	$\sim$0.35	$\sim$0.29
Untrained	`last_preaction`	L32	0.609	—	baseline

Untrained counterfactual comparison. Table 8 compares untrained counterfactual responses across architectures. VLM bases (Tier 1: Phi-3-Vision, Qwen2.5-VL) show weak sensitivity (cosine distance < 0.2), while OXE-pretrained models (Tier 2) show moderate sensitivity. The ordering reversal between trained (wrong > shuffled) and untrained VLM (shuffled > wrong) suggests training induces semantic rather than positional sensitivity.

4.2.3 L1 Attention Routing: Base-Model-Dependent

Table 9 presents L1 instruction-dominant head counts across architectures. The pattern is clear: L1 routing reflects the base model's pretraining, *not* VLA fine-tuning. TracVLA's untrained model (Phi-3-Vision) already has 32/32 instruction-dominant heads; training *reduces* this to 27–30/32 as some heads are repurposed. NORA shows stability (13–14/16 untrained vs. 12–13/16 trained). Only OpenVLA with random initialization (0/32) shows apparent training-induced L1 routing, but OXE-pretrained OpenVLA already has 25/32, confirming the effect originates from OXE pretraining, not task fine-tuning.

OpenVLA L1 attention detail. For the OpenVLA trained model, 24/32 heads are instruction-dominant with a mean text attention fraction of 57.7%, compared to 12% for untrained (random init) and 36.5% for Llama-2 base. The attention gradient—0/32 $\rightarrow$ 9/32 (Llama-2 base) $\rightarrow$ 24/32 (VLA trained)—tracks the progressive accumulation of language processing capability through pretraining stages, but the OXE comparison (25/32 untrained OXE $\rightarrow$ 24/32 trained) shows this is not a fine-tuning effect.

4.3 Three-Tier Routing Gradient

Table 10 synthesizes the three-tier gradient that emerges from comparing VLM bases (Tier 1), OXE-pretrained VLAs (Tier 2), and task-finetuned VLAs (Tier 3).

The gradient reveals that routing is not created wholesale by task fine-tuning. OXE pretraining on 970K diverse episodes already establishes strong L1 routing (25/32 heads) and substantial counterfactual sensitivity (d=1.05) without exit collapse (Δ=0.020). Task-specific fine-tuning then *reshapes* this existing routing: it induces exit collapse (action-relevant information selectively dropped at exit layers) and specializes counterfactual patterns to the target task, while L1 head counts remain approximately constant.

OpenVLA Goal three-tier detail. On LIBERO-Goal, the gradient is particularly clear (Table 11): Tier 1 (random init) has no routing capability (0/32 L1 heads, pathological R^2 collapse at L10–L13), Tier 2 (OXE) has established routing (25/32 heads, d=1.05, no exit collapse), and Tier 3 (task-finetuned) has specialized routing (24/32 heads, d=0.96, exit collapse Δ=0.150).

Table 6: Cross-benchmark R^2 peak comparison (Goal vs. Spatial).

Benchmark	Trained `img_mean` Peak	Trained `lastpre` Peak	Peak Shift
LIBERO-Goal	L8 (0.655)	L13 (0.678)	—
LIBERO-Spatial	L13 (0.696)	L21 (0.675)	+5 / +8 layers

Table 7: Counterfactual sensitivity: OpenVLA trained vs. untrained (OXE) on Goal and Spatial (Cohen's d for trained, cosine distance for untrained).

Suite	Condition	empty	wrong	shuffled	para
Goal	Trained	d=0.96	d=0.91	d=0.77	d=0.70
	OXE untr.	d=1.05	d=0.84	d=0.63	d=0.61
Spatial	Trained	d=0.94	d=0.85	d=0.83	d=0.81
	OXE untr.	d=1.02	d=0.82	d=0.61	d=0.59

4.4 Vision Encoder Geometry

4.4.1 Encoder-to-VLA RSA Comparison

Table 12 shows RSA between standalone encoders and action-aligned reference representations. Under RSA, the frozen SigLIP+projector retains the strongest geometric alignment (ρ=0.615, 95% CI [0.526, 0.695]). LLM processing degrades this alignment: the VLA trained peak (ρ=0.488) and untrained peak (ρ=0.485) are nearly identical, with a degradation magnitude of 0.127. DINOv2 cls_token (ρ=0.388) serves as a vision-only baseline.

4.4.2 Per-Layer RSA Profile

Table 13 shows the per-layer RSA profile. Trained models exhibit an inverted-U: RSA rises from L0 (0.612) to a peak at L12 (0.628, Δ=+0.016) before declining to L32 (0.422, Δ=−0.206). Untrained models show monotonic decay from L0 (0.612) to L32 (0.545). Both conditions share the same L0 starting point because they share a frozen projector. The projector-to-L1 transition produces the largest single drop (0.13–0.25), corresponding to the embedding dimension transformation.

4.4.3 Instruction Invariance of RSA

The 2×2 training × instruction matrix (Table 14) confirms that RSA is invariant to both training and instruction conditions, converging within a 0.48–0.49 band (bootstrap Δ CI includes 0). Extended to four instruction conditions (normal, random words, shuffled, empty), the maximum absolute deviation is 0.017, establishing triple invariance: RSA measures purely visual structure.

4.4.4 Cross-Encoder Comparison

Table 15 compares RSA and R^2 across encoder types. Under RSA, DINOv2 cls (ρ=0.692) outranks SigLIP+Proj (ρ=0.615) and CLIP ViT-L (ρ=0.557), suggesting that self-supervised spatial features provide stronger action-aligned geometry than vision-language contrastive features.

Branch-split decomposition. Within the fused backbone (DINOv2 + SigLIP), branch-split RSA (Table 16) shows equal contribution from both branches (0.619 vs. 0.621), with fusion providing a complementary gain (+0.015) and the projector introducing a small compression loss (−0.019).

4.4.5 Metric Robustness: RSA vs. CKA

Table 17 reveals an important metric dissociation. CKA yields an ordering reversal: untrained > trained > projector > DINO, while RSA gives projector > trained ≈ untrained > DINO. DINO ranks lowest under both

Table 8: Untrained counterfactual responses (cosine distance). Tier 1 = VLM base; Tier 2 = OXE pretrained. * denotes reference values from other cells.

Model	Suite	empty	shuffled	wrong	para	Tier
OXE (Llama-2)	Object	0.085	0.045	0.052	0.025	2
Phi-3-Vision	Spatial	0.148	0.078	0.018	0.005	1
Phi-3-Vision	Object	0.142	0.072	0.015	0.004	1
Qwen2.5-VL	Goal*	0.081	0.033	0.014	0.004	1

Table 9: L1 instruction-dominant heads: trained vs. untrained across architectures. L1 routing is base-model-dependent, not training-induced.

Architecture	Trained	Untrained	Training-Induced?	Source
OpenVLA (random init)	24/32	0/32	Appears so	No LLM pretraining
OpenVLA (OXE pretr.)	24/32	25/32	No	OXE pretraining
TracVLA	27–30/32	32/32	No (decreased)	Phi-3 pretraining
NORA	12–13/16	13–14/16	No (stable)	Qwen2.5-VL pretraining

metrics—a robust finding. The dissociation implies that LLM training degrades local geometric alignment (RSA) while preserving or enhancing global structure (CKA). This "geometric degradation + structural preservation" is richer than either metric alone would suggest, and all geometric claims in this paper are explicitly qualified by metric.

4.4.6 Weight Analysis and Frozen Encoder Verification

The vision encoder and projector remain perfectly frozen during training (cosine similarity = 1.0 and 1.001, respectively). LLM layers show uniform relative weight change (1.22–1.37×) across all 32 layers, with the output head at only 1.12×—ruling out shallow adaptation (threshold: 3×). This confirms that training-induced effects emerge from distributed LLM-wide reorganization, not localized head or layer modifications.

5 Discussion

What VLA training does and does not do. Our cross-architecture analysis reveals a clean separation between training-induced effects and inherited properties. VLA fine-tuning *does* create R^2 exit collapse and enhance counterfactual sensitivity—these are architecture-independent signatures observable across Llama-2-7B, Phi-3-mini, and Qwen2.5-VL-3B backbones. VLA fine-tuning does *not* create L1 attention routing: this is inherited from the base model's pretraining (language modeling, vision-language alignment, or OXE pretraining), and task fine-tuning may even reduce the number of instruction-dominant heads.

The routing gradient. The three-tier gradient—VLM base → OXE VLA → task VLA—suggests that instruction routing is progressively refined rather than created wholesale. OXE pretraining establishes general-purpose routing (L1 heads, counterfactual sensitivity) without task specialization (no exit collapse). Task fine-tuning then reshapes this routing, concentrating action-relevant information in mid-layers while allowing exit layers to specialize for action output. This distinction has practical implications: diagnostic tools can use exit collapse as a signature that task-specific routing has been learned.

Metric dependence of geometric findings. Our RSA and CKA results diverge on encoder rankings and LLM-layer trajectories. RSA captures local geometric alignment (which LLM processing degrades), while CKA captures global structural preservation (which LLM processing maintains). We present this not as a limitation but as a richer characterization: VLA training simultaneously degrades fine-grained geometric alignment and preserves coarse structural organization. All geometric claims in this paper are explicitly qualified by the metric used.

Table 10: Three-tier routing gradient. Routing emerges at the VLM→VLA transition (Tier 1→2) and reshapes with task specialization (Tier 2→3).

Tier	Representative	Exit Collapse	CF (empty)	L1 Routing	Interp.
1: VLM base	Phi-3-Vision, Qwen2.5-VL	None ($\Delta < 0.05$)	$\cos < 0.2$	From LLM pretr.	No action routing
2: OXE VLA	openvla-7b (970K ep.)	None ($\Delta = 0.02$)	$d \approx 1.05$	25/32 from OXE	General routing
3: Task VLA	All 3 finetuned	Yes ($\Delta = 0.12$–0.69)	$d = 0.7$–1.5	$\approx$ Tier 2	Task-specific

Table 11: Three-tier routing gradient on LIBERO-Goal (OpenVLA).

Tier	Model	L1 Heads	CF d (empty)	R^2 Exit Δ	Collapse?
1 (VLM base)	Llama-2-7B random init	0/32	$\sim$0	N/A (pathological)	N/A
2 (OXE VLA)	openvla-7b	25/32	1.05	0.020	No
3 (Task VLA)	openvla-goal-finetuned	24/32	0.96	0.150	Yes

Limitations. Our analysis is conducted in an open-loop setting without closed-loop robot evaluation, limiting claims about behavioral consequences. The LIBERO benchmark, while diverse, uses a fixed embodiment; generalization to other platforms remains untested. NORA's 2-of-4 suite coverage (no Object/Long checkpoints) leaves gaps in the matrix. The OFT outlier (OpenVLA Long, Δ=0.004) suggests that parameter-efficient fine-tuning methods may circumvent the exit collapse signature, warranting further investigation.

6 Conclusion

We presented a systematic token-level analysis framework for VLAs and applied it to three architectures across four task suites. Our central finding is that VLA fine-tuning produces two architecture-independent training signatures—R^2 exit collapse and heightened counterfactual sensitivity—confirmed in 9 of 10 matrix cells across three VLA families. A three-tier routing gradient reveals that instruction routing emerges at the VLM-to-VLA transition and reshapes with task specialization, with L1 attention routing inherited from the base model rather than created by fine-tuning. These findings provide mechanistic grounding for VLA design: entry-exit routing patterns and exit collapse diagnostics can guide architecture optimization and training monitoring.

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Table 12: Encoder-to-VLA RSA comparison. LLM processing degrades geometric alignment from the frozen encoder; training does not recover it (trained ≈ untrained). DINOv2 establishes a vision-only floor.

Source	RSA ρ	95% CI	vs. DINO Δ
SigLIP+Proj	0.615	[0.526, 0.695]	+0.227
VLA Trained Peak	0.488	—	+0.100
VLA Untrained Peak	0.485	—	+0.097
DINO cls_token	0.388	—	baseline

Table 13: Per-layer RSA profile summary (OpenVLA, LIBERO-Goal, `image_mean`).

Condition	L0	Peak Layer	Peak RSA	L32	Δ(peak–exit)
Trained `img_mean`	0.612	L12	0.628	0.422	0.206
Untrained `img_mean`	0.612	L0	0.612	0.545	0.067

Table 14: 2×2 RSA convergence matrix. All four conditions converge within [0.48, 0.49]; neither training nor instruction presence has a significant effect (bootstrap CI includes 0).

	With Instruction	No Instruction	Instruction Effect
Trained	0.488	0.489	−0.001
Untrained	0.485	0.480	+0.004
Training Effect	+0.003	+0.009	

Table 15: Cross-encoder comparison (RSA and action-probe R^2).

Encoder	RSA	R^2 (action)	Ranking
DINOv2 cls	0.692 / 0.631	0.634	1st / 2nd
SigLIP+Proj	0.615	0.630	2nd / 3rd
CLIP ViT-L	0.557	0.587	4th / 5th

Table 16: Branch-split RSA decomposition of the fused backbone.

Component	RSA	Note
DINOv2 branch (1088d)	0.619	in-backbone
SigLIP branch (1088d)	0.621	in-backbone
Fused (2176d)	0.634	complementary gain
Post-projector (4096d)	0.615	compression loss −0.019

Table 17: RSA vs. CKA ordering comparison. Ordering reversal between metrics reflects different aspects of representational structure.

Source	RSA Rank	Linear CKA Rank	RDM-CKA Rank
SigLIP+Proj	1st (0.615)	3rd (0.081)	2nd (0.257)
VLA Trained	2nd (0.488)	4th (0.083)	1st (0.260)
VLA Untrained	3rd (0.485)	2nd (0.089)	3rd (0.246)
DINO	4th (0.388)	5th (0.066)	4th (0.241)